SKILL BUILDERS · UNIT 2 · SCORE 3

Exponents & Scientific Notation: Practice Set

NC.8.EE.1, 3–4

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.EE.1

Simplify: $4^3 \times 4^2$

(A) 16^5 (B) **4^5**

(C) 4^6 (D) 4^1

2 Score 3 NC.8.EE.1

Simplify: $x^7 \div x^4$

(A) $x^{7/4}$ (B) x^{11}

(C) x^{28} (D) **x^3**

3 Score 3 NC.8.EE.1

Simplify: $(2^3)^2$

(A) **2^6** (B) 4^3

(C) 2^9 (D) 2^5

4 Score 3 NC.8.EE.1

Find the value of 5^0.

Answer: **1**

5 Score 3 NC.8.EE.1

Find the value of 3^{-2}. (Type a fraction like 1/4.)

Answer: **1/9**

1. $3^{-2} = 1/3^2$
2. $= 1/9$

6 Score 3 NC.8.EE.1

Find the value of $(-3)^2$.

Answer: **9**

1. $(-3)^2 = (-3)(-3)$
2. $= 9$

7 Score 3 NC.8.EE.3

Write 0.00056 in scientific notation.

(A) 5.6×10^{-3} (B) **5.6×10^{-4}**

(C) 56×10^{-5} (D) 5.6×10^4

8 Score 3 NC.8.EE.3

Write 7.2×10^6 in standard form.

Answer: **7200000**

1. 7.2×10^6: move the point 6 places right
2. $= 7{,}200{,}000$

9 Score 3 NC.8.EE.4

Multiply: $(5 \times 10^3)(3 \times 10^4)$

- **A** 1.5×10^8
- B 1.5×10^7
- C 15×10^7
- D 8×10^7

10 Score 3 NC.8.EE.4

Divide: $(8 \times 10^7) \div (4 \times 10^3)$

- A 2×10^{10}
- **B** 2×10^4
- C 4×10^4
- D 32×10^4

11 Score 3 NC.8.EE.1

Find the mistake. Lee wrote: $x^3 \cdot x^4 = x^{12}$. What went wrong?

- **A** **Lee multiplied the exponents. Multiplying powers adds them: x^7.**
- B Lee is right.
- C It should be $2x^7$.
- D It should be x^1 because $4 - 3 = 1$.